



- (e) The 8-bit binary numbers 01100101 and 01110000 are stored in RAM.

The binary numbers are added together.

Add the **two** binary numbers using binary addition.

Give your answer in binary. You must show all your working.

$$\begin{array}{r} 01100101 \\ + 01110000 \\ \hline \end{array}$$

[3]

- (f) An overflow occurs when two other 8-bit binary numbers are added together.

Explain why the overflow error occurs.

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..... [2]

- (g) The negative denary number –22 is stored in RAM.

Negative denary numbers can be represented as binary using two's complement.

Give the two's complement 8-bit binary integer that would be stored for the denary number –22.

You must show all your working.

Working space

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Two's complement 8-bit binary integer [2]

